

MFG-04-QUALITY-CONTROL

MFG-04: Carrington Storm Motors — Quality Control and Assurance

Document ID: CSM/MFG/04 **Version:** 1.0 — Investor-Grade **Classification:** Confidential **Heuristic Foundation:** Accountant (quality metrics discipline), Zirconia (materials testing rigor), El Segundo (zero-failure imperative) **Agents Consulted:** Manufacturing domain agent, Zirconia, Quality domain agent, Accountant **Date:** July 2026

1.0 EXECUTIVE SUMMARY

For Carrington Storm Motors, quality is not a cost center — it is the product. A transformer protection system that fails during a Carrington Event doesn’t just lose a customer — it potentially causes a cascading grid failure that could have been prevented. The consequences of quality failure are measured in billions of dollars and thousands of lives.

This document outlines CSM’s quality control and quality assurance framework: incoming material inspection, in-process quality checks, final product testing, statistical process control, Six Sigma program, ISO 9001/AS9100 certification timeline, warranty/returns policy, and continuous improvement program.

As Zirconia frames it: “When the Sun throws its worst at the grid, your quality control program is the only thing between a surviving transformer and a burning one. Quality is not a department. Quality is the mission.”

2.0 INCOMING MATERIAL INSPECTION

2.1 Critical Material Inspection Protocols

Material	Key Properties Tested	Test Method	Frequency	Acceptance Criteria
MAX phase (Ti3AlC2)	Phase purity (>95% MAX phase); particle size distribution; elemental composition	XRD; SEM/EDS; laser diffraction	Every lot	Per CSM-MAT-001 specification
LiF	Purity (>99%); moisture content	Titration; Karl Fischer	Every lot	Per ACS reagent grade
Basalt fiber roving	Tensile strength; tex (linear density); sizing compatibility	Single-filament tensile test (ASTM D2101); gravimetric	Every lot	Per CSM-MAT-004
CNTs (MWCNT)	Diameter distribution; length; purity (TGA residue); conductivity	TEM; TGA/DSC; four-point probe	Every lot	Per CSM-MAT-007
Epoxy resin	Viscosity; epoxy equivalent weight; gel time	Rheometer; titration; DSC	Every lot	Per manufacturer CoA + CSM verification
BaTiO3 powder	Particle size; tetragonality (c/a ratio); purity	Laser diffraction; XRD; XRF	Every lot	Per CSM-MAT-009
Metakaolin	Reactive alumina/silica content; particle size; LOI	XRF; laser diffraction; TGA	Every lot	Per CSM-MAT-011

2.2 Supplier Certificate of Analysis (CoA) Requirements

Every material shipment must be accompanied by a CoA documenting: - Batch/lot number for traceability - Test results for all specified properties - Test methods used - Acceptance criteria - Date of manufacture - Expiration date (if applicable) - Signature of supplier’s quality representative

CSM verifies CoA results through its own incoming inspection on a statistical sampling basis (ANSI/ASQ Z1.4, AQL 1.0 for critical properties, AQL 2.5 for non-critical).

3.0 IN-PROCESS QUALITY CHECKS

3.1 MXene Processing In-Process Controls

Process Step	In-Process Check	Method	Frequency	Specification
Etching — MAX phase to MXene	Etch completion (Al removal >95%)	XRD (disappearance of MAX phase peaks)	Every batch	104 peak absent in XRD
Delamination	Delamination yield; sheet size distribution	Centrifugation yield; dynamic light scattering (DLS)	Every batch	>80% delamination yield; median flake size 2-5 µm
MXene ink formulation	Viscosity; solid content; sedimentation stability	Rheometer; gravimetric; settling test	Every batch	Per CSM-PROC-001
Film casting/drying	Wet film thickness; dry film thickness; visual defects	Thickness gauge; automated optical inspection (AOI)	Continuous (in-line)	Thickness ±10% of target; <1 defect/cm²
Sheet resistance	Four-point probe measurement	In-line or off-line	Every panel	<10 Ω/sq (for standard shielding grade)

3.2 BFRP Processing In-Process Controls

Process Step	In-Process Check	Method	Frequency	Specification
Fiber tension	Roving tension	Tensiometer	Continuous	Per CSM-PROC-004
Resin bath	Resin viscosity; temperature; pot life	Rheometer; thermocouple; timer	Every 4 hours	Within manufacturer's process window
Die temperature	Multiple zone temperatures	Thermocouples (PLC-monitored)	Continuous	Per CSM-PROC-004 (profile-specific)
Pull speed	Line speed	Tachometer	Continuous	Per product specification
Profile dimensions	Cross-section dimensions; straightness	Calipers; straightness gauge	Every 10 meters	Per CSM-SPEC-010

3.3 General In-Process Controls

Control	All Products	Method
Environmental monitoring	MXene processing area: temperature 20±2°C, humidity <30% RH	Environmental sensors; data-logged
Clean room particle count	ISO 7 (Class 10,000): <352,000 particles/m³ at 0.5µm	Particle counter; daily
Equipment calibration	All measurement equipment	Per calibration schedule (ISO 17025 traceable)
Tooling condition	Molds, dies, cutting tools	Visual inspection; dimensional check; replacement per life count

4.0 FINAL PRODUCT TESTING

4.1 Aegis Home — Final Acceptance Test

Test	Method	Acceptance Criteria	Sampling
Visual inspection	AOI + human inspector	No visible defects; labels correct; packaging intact	100%
Shielding effectiveness	Custom TEM cell (production-qualified)	SE >90 dB at 10 GHz (production spec; >92 dB design target)	100%
GIC blocking	Simulated GIC injection	>94% GIC reduction (standard); >98% (enhanced)	100%
Dielectric withstand (HiPot)	High-potential test per UL 1449	No breakdown at 1.5x rated voltage	100%
Ground continuity	Resistance measurement	<0.1 Ω between grounding terminals	100%
Functional test — indicators	All status LEDs, audio, tactile indicators verified	All operational	100%
Functional test — connectivity	WiFi/Bluetooth connectivity test (monitoring version)	Connects to test network; data transmission verified	Per monitoring-equipped units: 100%
Packaging integrity	Drop test; vibration test	Per ISTA 3A (packaging performance)	Per lot (statistical)

4.2 Charlemagne Fleet — Final Acceptance Test

Test	Method	Acceptance Criteria	Sampling
Component-level shielding test	TEM cell (large-format)	SE >90 dB	100% of shielding panels
GIC blocking device test	Full-current GIC simulation (up to 500A)	>99.9% GIC reduction at rated current	100%
High-voltage test	Per IEEE C57 (transformer testing)	No breakdown at test voltage	100%
System integration test	Connected system test with simulated transformer	All components function as integrated system	Each installation kit
Documentation review	Installation manual; test certificate; warranty registration	Complete and accurate	100%

4.3 Product Life Testing (Accelerated Aging)

Test	Conditions	Duration (Accelerated)	Equivalent Service Life	Sampling
Temperature cycling	-40°C to +85°C, 4-hour cycles	1,000 cycles (~6 months)	20 years	Per product family; initial lot
Humidity exposure	85°C / 85% RH	1,000 hours	10-15 years	Per product family
UV exposure	Xenon arc (ASTM G155)	2,000 hours	20 years outdoor equivalent	BFRP components
Salt fog (corrosion)	ASTM B117	1,000 hours	20 years coastal exposure	External components

Vibration	Random vibration per MIL-STD-810	Per test standard	20 year transportation/operational	Per product
Mechanical shock	Drop test; impact test	Per ISTA standards	Transportation lifetime	Per product
Electrical endurance	Repeated GIC injection (1,000 cycles)	1,000 cycles at rated current	100+ Carrington-level events	Per GIC blocking device design

5.0 STATISTICAL PROCESS CONTROL (SPC)

5.1 SPC Implementation

Process	Key Characteristic (SPC Chart)	Control Chart Type	Monitoring
MXene film thickness	Dry thickness (µm)	X-bar and R chart	Per production run
MXene sheet resistance	Ω/sq	X-bar and R chart	Per production run
BFRP profile dimension	Width, thickness (mm)	X-bar and R chart	Continuous (in-line measurement)
BFRP tensile strength	MPa	X-bar and R chart	Per lot
Ceramic density	g/cm³	X-bar and R chart	Per sintering batch
Geopolymer compressive strength	MPa (7-day and 28-day)	X-bar and R chart	Per mix batch
Assembly cycle time	Minutes per unit	X-bar and R chart	Daily
First-pass yield	% pass	p-chart (fraction nonconforming)	Daily
Customer complaints	Count per month	c-chart	Monthly

5.2 Process Capability Targets

Process	Capability Target (Cpk)	Notes
Critical safety characteristics (shielding SE, GIC reduction, HiPot)	Cpk > 1.67	Six Sigma — safety-critical
Key performance characteristics (sheet resistance, mechanical strength)	Cpk > 1.33	Four Sigma
Non-critical characteristics (cosmetic, packaging)	Cpk > 1.0	Three Sigma

6.0 SIX SIGMA PROGRAM

6.1 Six Sigma Deployment

Belt Level	Headcount (Phase 2)	Headcount (Phase 3)	Role
			Program leadership;

Master Black Belt	1 (external consultant initially)	1-2 (internal)	training; strategic projects Full-time improvement project leaders Part-time project leaders (in addition to regular role) Basic problem- solving training (all engineers and supervisors)
Black Belt	1-2	3-5	
Green Belt	5-10	20-50	
Yellow Belt	20-50	100-300	

6.2 Six Sigma Project Selection

Priority projects are selected based on: - **Safety impact:** Projects reducing risk of product failure during GMD events - **Quality impact:** Projects improving first-pass yield and reducing customer complaints - **Cost impact:** Projects reducing manufacturing cost (waste, rework, efficiency) - **Customer impact:** Projects improving customer experience (installation ease, monitoring reliability)

Target: 2-4 Black Belt projects and 10-20 Green Belt projects active at any time (Phase 2).

7.0 ISO CERTIFICATION TIMELINE

7.1 ISO 9001:2015 — Quality Management Systems

Phase	Activity	Timeline
1. Gap Analysis	Assess current state against ISO 9001 requirements	Year 1, Month 6
2. Documentation	Develop quality manual, procedures, work instructions, forms	Year 1, Months 6-12
3. Implementation	Implement QMS across pilot facility	Year 1, Month 12 — Year 2, Month 6
4. Internal Audit	Train internal auditors; conduct first internal audit	Year 2, Month 6
5. Management Review	First management review of QMS	Year 2, Month 9
6. Certification Audit	Stage 1 audit (documentation review)	Year 2, Month 9-10
7. Certification Audit	Stage 2 audit (implementation verification)	Year 2, Month 11-12
8. Certification	ISO 9001:2015 certificate issued	Year 2, Month 12 — Year 3, Month 3

7.2 AS9100D — Aerospace/Defense Quality Management

AS9100 adds ISO 9001 requirements for aerospace and defense: risk management, configuration management, foreign object debris (FOD) prevention, and additional design/development controls. Required for DOD contracts.

Phase	Timeline
AS9100 implementation	Year 3-4 (after ISO 9001 achieved)
AS9100 certification	Year 4-5

7.3 ISO 14001 and ISO 45001

Standard	Timeline
ISO 14001 (Environmental Management)	Year 2-3
ISO 45001 (Occupational Health and Safety)	Year 3-4

8.0 WARRANTY AND RETURNS POLICY

8.1 Warranty Terms

Product	Warranty Period	Coverage
Aegis Home	10 years (standard); lifetime (enhanced)	Defects in materials and workmanship; failure to meet published performance specifications
Aegis Business	10 years	Defects in materials and workmanship
Charlemagne Fleet	20 years	Defects in materials and workmanship; GIC reduction performance guarantee
CSM Maritime/Fleet	10 years	Defects in materials and workmanship

8.2 Warranty Reserve Accounting

CSM accrues a warranty reserve based on estimated future warranty costs as products are sold (per ASC 450 / FAS 5). Initial reserve rate: 2-3% of product revenue (to be adjusted based on actual experience).

8.3 Returns and RMA Process

Step	Timeline	Activity
1. Claim Initiation	Day 0	Customer contacts CSM; RMA (Return Material Authorization) number issued
2. Triage	Day 1-2	CSM technical support assesses issue; remote diagnostics (monitoring-equipped units)
3. Return (if needed)	Day 3-7	Prepaid return shipping label; unit returned to CSM
4. Analysis	Day 8-21	Root cause analysis; failure mode determination
5. Resolution	Day 22-30	Repair, replacement, or refund; customer communication
6. CAPA (Corrective Action)	Day 31-60	If significant failure: CAPA initiated; root cause addressed in manufacturing

9.0 CONTINUOUS IMPROVEMENT (KAIZEN)

9.1 Kaizen Program Structure

Element	Description	Frequency
Daily stand-up (cell level)	10-minute team meeting: yesterday’s issues, today’s priorities, improvement ideas	Daily
Kaizen event	3-5 day focused improvement workshop on specific process	Monthly (rotating across production lines)
Gemba walk	Leadership walks the production floor, observes, asks questions	Weekly (plant manager); monthly (CEO/COO)

Suggestion system	All employees submit improvement ideas; review committee evaluates; implemented ideas rewarded	Ongoing (monthly review)
A3 problem solving	Structured problem-solving methodology for complex issues	As needed

9.2 Continuous Improvement Metrics

Metric	Phase 1 Baseline	Phase 2 Target	Phase 3 Target
Kaizen events per year	0 (not yet)	12+	50+
Employee suggestions per year	N/A	1 per employee	2+ per employee
% suggestions implemented	N/A	>40%	>50%
Cost savings from improvement (annual)	N/A	3-5% of manufacturing cost	5-8%
First-pass yield improvement (year-over-year)	N/A	+5-10 percentage points	+2-5 percentage points
Customer complaint reduction (YoY)	N/A	-30%	-20%

10.0 NON-CONFORMANCE AND CAPA

10.1 Non-Conformance Process

Step	Description
Identification	Any employee can identify a non-conformance (product, process, system)
Documentation	NCR (Non-Conformance Report) created in QMS
Containment	Immediate action to isolate non-conforming product; prevent further production
Disposition	Decision: rework, scrap, accept with concession (customer approval required), return to supplier
Root Cause Analysis	5 Why, Fishbone (Ishikawa), or FMEA methodology
Corrective Action	Action to eliminate root cause
Preventive Action	Action to prevent recurrence in similar products/processes
Verification	Verify corrective action is effective (audit, data review)
Closure	NCR closed when all actions complete and verified

10.2 CAPA (Corrective and Preventive Action) System

CAPA is the systematic approach to investigating and correcting quality issues: - **Level 1 CAPA:** Minor issue; resolved at production cell level within 48 hours - **Level 2 CAPA:** Significant issue; cross-functional team; 30-day resolution target - **Level 3 CAPA:** Critical issue (safety, regulatory, major customer impact); executive escalation; 72-hour initial containment; 60-day full resolution

11.0 AGENT VOICES

Zirconia (Testing Rigor): “Every Aegis Home that leaves the factory must be tested for shielding effectiveness. Not a sample — every unit. The cost of 100% testing is a few dollars per unit in amortized equipment and cycle

time. The cost of one untested unit failing during a Carrington Event is a destroyed home, a destroyed reputation, and a class-action lawsuit. 100% testing is the cheapest insurance CSM will ever buy.”

Quality Domain Agent: “The accelerated life testing program is the most important quality investment CSM will make. We need to simulate 20 years of service in 6 months of testing — temperature cycling, humidity, UV, vibration, electrical cycling. If MXene oxidizes, if BFRP delaminates, if ceramics crack — we need to find out in the lab, not in the field. The testing results will determine the warranty terms, the product design iterations, and the customer confidence.”

The Accountant (Quality Economics): “Quality costs money — prevention costs (training, SPC, calibration), appraisal costs (inspection, testing), and failure costs (internal — scrap/rework; external — warranty/liability). The goal is not zero quality cost — it’s optimizing total quality cost. Investing \$1 in prevention typically saves \$10 in appraisal and \$100 in failure cost. The Accountant Heuristic says: measure all three categories of quality cost. Optimize the total. The right answer is not ‘spend as little as possible on quality’ — it’s ‘spend the right amount so that total quality cost is minimized.’ For CSM, given the consequences of failure, ‘the right amount’ is substantially higher than for most manufacturers.”

Mork (Quality Reality): “Quality programs are paperwork exercises until something goes wrong. Then they’re either the thing that caught the problem before it reached the customer — or the thing that should have caught it and didn’t. CSM’s quality culture must be obsessed with finding problems internally, before customers find them externally. Celebrate people who find defects, don’t punish them. Reward the person who shut down the production line because something looked wrong. That person just saved the company millions and maybe saved lives. That’s the culture we need.”

Document prepared for investor and operational planning purposes. All quality control specifications, testing protocols, and certification timelines are targets that will be refined through implementation experience. Quality management system development should be led by an experienced Quality Director hired in Year 1.